MIL-E-1/335A
20 February 1970
SUPERSEDING
MIL-E-1/335
14 August 1953

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRANSMITTING

TYPE 2E22

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, rf power amplifier, oscillator, modulator

Outline	—	See figure 1
Cap	—	Medium with ceramic insulator
Base	—	A5-11 (low-loss phenolic)
Envelope	—	ST16
Cathode	—	Coated filament
Mounting	—	See note 1

Base connections:

Pin No.	—	1	2	3	4	5	Cap
Element (see note 2)	—	f	g2	g1	g3	f	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ib	Ic1	Pp	Pg2	Pi	Modulation	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	W	W	W	—	ft
C Telep:												
Maximum:	6.9	750	-200	275	-200	60	9.0	33	10	45	Suppressor	10,000
Minimum:	5.7	—	—	—	—	—	—	—	—	—	—	—
Maximum:	6.9	600	-200	275	0	70	9.0	21	10	55	Anode	10,000
Minimum:	5.7	—	—	—	—	—	—	—	—	—	—	—
C Teleg:												
Maximum:	6.9	750	-200	275	0	110	9.0	33	10	83	—	10,000
Minimum:	5.7	—	—	—	—	—	—	—	—	—	—	—
TEST CONDITIONS:	6.3	500	-15	250	0	—	—	—	—	—	—	—

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	See note 3	1.0	II	Ic1	0	-5.0	µAdc
1236	Power oscillation	Ef = 5.5 Vac; F = 15 MHz; Eb = 750 Vdc; Rg = 10,000 ohms; Esig/Ic1 = 5 mAdc; Ib = 100 mAdc	1.0	II	Po	43.0	---	W
1231	Emission	Eb = Ec1 = Ec2 = Ec3 = 50 Vdc (see note 3)	1.0	II	Is	250	---	mAdc
1256	Electrode current (screen grid)		1.0	II	Ic2	0.5	6.5	mAdc
1256	Electrode current (suppressor grid)	Ec3 = 45 Vdc	1.0	II	Ic3	---	-400	µAdc
1256	Electrode current (1) (anode)		1.0	II	Ib	42.0	78.0	mAdc
1256	Electrode current (2) (anode)	Ec3 = -170 Vdc	1.0	II	Ib	0.5	10.0	mAdc
(A) 1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Filament current		---	---	If	1.35	1.65	A
1266	Grid emission	Ec3 = 45 Vdc (see note 4)	---	---	---	---	---	---
1311	Anode resistance	Ec1 = -12 Vdc	---	---	rp	80,000	---	Ohms
1306	Transconductance	Ec1 = -12 Vdc	---	---	Sm	4,200	6,800	µmhos
---	Operation peak output voltage	Ebb = Ecc2 = 500 Vdc; Ec3 = -30 Vdc; RL = 1,500 ohms; Rg2 = 10,000 ohms; R2 = 30,000 ohms; Eg1 = 45 Vac; 60 Hz; Rg1 = 10,000 ohms; C1 = C2 = 40 µF (see figure 2)	---	---	ep	250	---	v
1331	Direct-interelectrode capacitance	Shield No. 312	---	---	Cgp	---	0.2	pF
		Without shield			Cin	11.5	15.5	pF
		Without shield			Cout	5.8	9.3	pF
1031	Low-frequency vibration	Eb = 250 Vdc; Ec = -12 Vdc; Rp = 2,000 ohms; Ec2 = 180 Vdc	---	---	Ep	---	500	mVac
1036	Bump	Angle = 20°	---	---	---	---	---	---
(A) 1101	Secureness of base, cap, or insert		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
(A) 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; Eb = 750 Vdc; Ec/Ib = 40 mAdc	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours)	Emission	---	---	Is	200	---	mAdc

NOTES:

1. The tube may be used in horizontal position only when pins 2 and 4 are in vertical plane.
2. The plane of the filament assembly shall not vary more than ±22-1/2° with the plane passing through pins 2 and 4 and parallel to the axis of the tube.
3. This test to be performed at the conclusion of the holding period.
4. (A) Return suppressor grid to filament potential through 100,000-ohm resistor. Note anode current. Apply positive suppressor voltage momentarily across the 100,000-ohm resistor. Upon removal of positive suppressor voltage, the anode current shall return immediately to its initial value within ±1 milliampere. Repeat test three times.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 11, 70, 85
DSA - ES

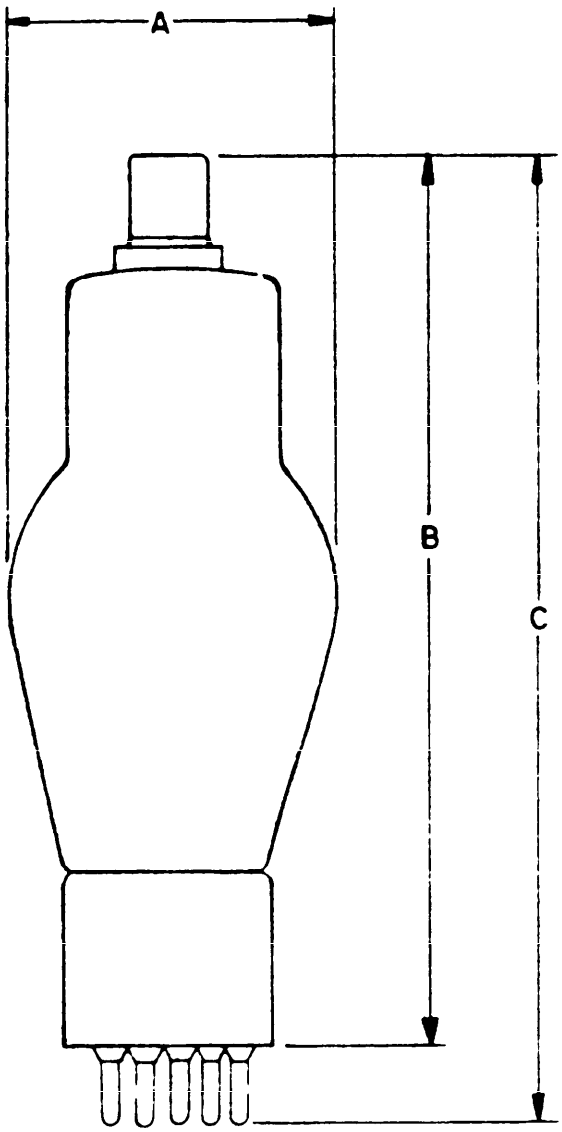
User activities:
Army - ME, AV
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2401-15)

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Dimensions in inches with metric equivalents (mm) in parentheses		
Ltr	Minimum	Maximum
Quality conformance inspection, part 1		
A	- - -	2.0625 (52.39)
B	- - -	5.500 (139.70)
C	5.8125 (147.64)	6.125 (155.58)

FIGURE 1. Outline drawing of electron tube type 2E22.

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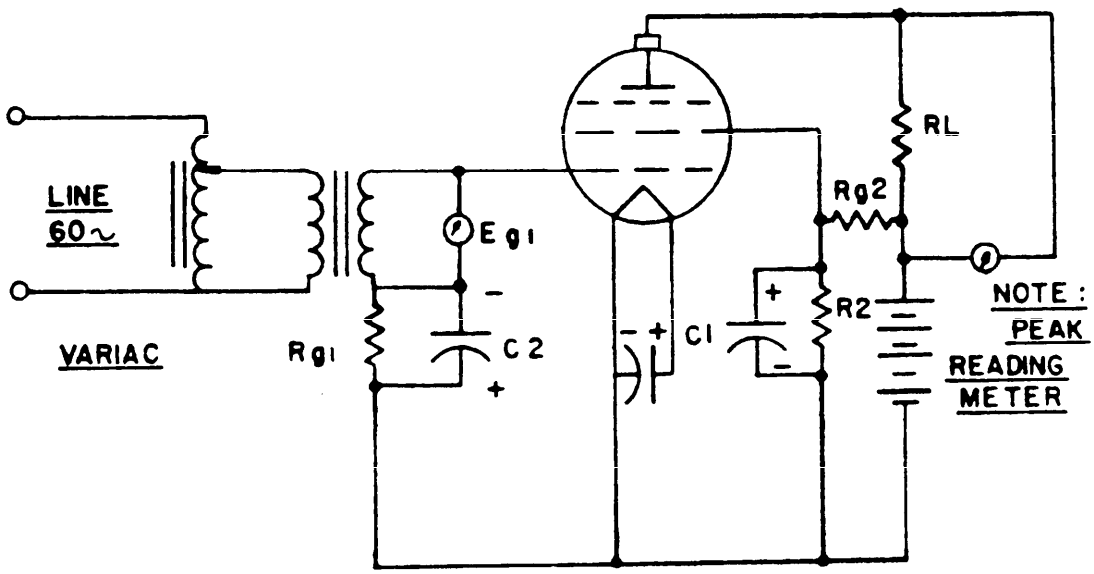


FIGURE 2. Circuit for peak output voltage test.

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WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 83 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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